Debendra Prasad Sahoo

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EDUCATION

Indian Institute of Technology, Mandi : MBA (Data Science & AI) (CGPA: **7.7/10**) 2025–present
Odisha University of Technology and Research : B.Sc. Physics (CGPA: **7.56/10**) 2022–2025
CBSE (10th: 91.6%, 12th: 91.8%)

EXPERIENCE

Artificial Intelligence Intern, DST CoE in Climate Modelling, IIT Delhi — May 2024 – Jan 2025
Role: AI Intern — *New Delhi, India*

- Implemented regression-based ensemble model for climate prediction, achieving **18% improvement** over baselines.
- Managed data preprocessing, feature engineering, and developed HPC pipelines; optimized for large datasets.
- Lead author on research paper on AI in climate science (in progress).
- Supported technical content and organization for Climate Change Conclave at IIT Delhi.

AI / ROS Intern, SakRobotix Lab Pvt. Ltd. — Jan 2024 – Feb 2024
Role: AI & ROS Intern — *Bhubaneswar, Odisha*

- Designed CNNs for Mel-spectrogram matching to improve wake-word detection.
- Benchmarked multiple pretrained audio models for voice recognition.
- Worked on lightweight model deployments for edge inference.

LEADERSHIP & RESPONSIBILITIES

Technical Lead – Aerospace Club, OUTR — 2024 – 2025
Large Scale Cross-Functional Team Management — *Bhubaneswar, Odisha*

- Led drone and RC plane projects; increased membership by 200%+ and coordinated cross-functional recruitment.
- Revived astronomy wing; organized sky-observation events; led autonomy and real-time data-analysis robotics initiatives.

Project Lead – Robotics & Autonomous Systems Initiative — 2022 – 2024
End-to-End Product & Business Development — *Bhubaneswar, Odisha*

- Directed the product lifecycle for autonomous drones and 3D-printed components, securing INR 3,00,000 in university funding for ISRO IRoC-U and delivering solutions to research clients to generate INR 4,00,000 in cash revenue.
- Scaled project to successfully qualifying for a 5L institutional incubation grant by demonstrating operational efficiency and market viability.

Project Lead – Robotics & Computer Vision — 2023 – 2024
Led nationally recognized initiatives — *Bhubaneswar, Odisha*

- **ISRO Rover Challenge (IRoC-U 2024)**: top 50/1400; built autonomous rover using ROS2 + MoveIt2.
- **e-Yantra IIT Bombay 2023–24**: developed ROS2 hologlyph bots; reached pre-final hardware stage.
- **Flipkart Grid 5.0**: engineered ROS + YOLO vision stack for robotic arm payload handling.

Management & Corporate Relations Lead – Team Deimos (Mars Rover) — 2025 – present
Technical Program Management & Strategic Partnerships — *Mandi, Himachal Pradesh*

- Managed cross-functional engineering timelines by implementing Work Breakdown Structures (WBS) and Gantt charts; optimized team workflows across GitHub and Notion.
- Spearheaded corporate outreach to secure project resources, successfully negotiating and advancing strategic partnerships with global brands like Creality, WOL3D and multiple others in pipeline.
- Managed Team Deimos's financial planning by developing and overseeing the master budget tables required for the international University Rover Challenge (URC) submission.

PROJECTS

- **FinTech Risk Modeling:** Engineered a predictive credit risk evaluation model for a Japanese small finance bank to mitigate default risks and optimize data-driven lending decisions.
- **Automated Content Creation AI Agent:** Architected a multi-agent system deployed on Azure (OpenAI/DALL-E) to fully automate and scale multimedia content generation for LinkedIn and Twitter.
- **Agri-Tech Predictive Analytics:** Developed a machine learning model (RandomForest) for the Smart India Hackathon to forecast crop water requirements and optimize resource allocation.
- **Applied GenAI in Healthcare:** Fine-tuned LLaMA 2 architectures for domain-specific medical applications to improve targeted information retrieval.
- **Autonomous UAV Vision Systems:** Directed the build and testing of multiple drone platforms, integrating real-time computer vision (object detection) for live-feed monitoring.
- **Algorithmic Optimization:** Implemented reinforcement learning models utilizing OpenAI Gymnasium for complex decision-making environments.

WORKSHOPS & MENTORING

Mentor – Skill++ 2024, Zairza Technical Club (OUTR) **2024**

Instructor & Mentor *Bhubaneswar, Odisha*

- Mentored 12 students on robotics fundamentals with ROS; guided TurtleBot simulation projects.
- Delivered sessions on robot perception with curated learning resources.

AWARDS & EXTRACURRICULAR

- Best Delegate at NITR IMUN 2023 (Represented EU at G20 Summit): 1st prize.
- Top ranks in robotics challenges (IRoC-U 47/1400 teams, Round 2 Stage 6 e-Yantra).
- 6-year member of SCAAA: promoted astronomy awareness and managed telescopes for public outreach.

SKILLS

Languages: Python, SQL, C++, C
Frameworks & Libraries: Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, OpenCV
Tools & Platforms: Git, ROS, GCP, Power BI, n8n, Notion, Excel, APIs
Domains: Machine Learning, NLP, Computer Vision, High-Performance Computing (HPC), Business Analytics, Finance
Professional: Cross-functional Leadership, Strategic Partnerships, Research Writing, Product/Project Management